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APPLIED RESEARCH

Research on YOLOv5s Improved Algorithm for Pavement Crack Detection in Complex Environments

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ABSTRACT Aiming to solve the problem of pavement crack detection in complex road environments, an improved algorithm based on YOLOv5s is proposed. First, the CBAM (Convolutional Block Attention Module) is introduced after the backbone network's C3 modules to focus the complex scene's practical information and enhance the model's attention to the crack region. Second, the BiFPN (Bidirectional Feature Pyramid Network) is used in Neck to replace the bidirectional PANet (Path Aggregation Network) in YOLOv5s to improve the multiscale feature fusion, which reduces detection leakage due to illumination and scale factors. Thirdly, the P6 detection head for denser cracks is added to the Head, and the CA (Coordinate Attention) module is introduced to improve the crack detection capability at multiple scales. Finally, the algorithm is experimentally compared with SSD (Single Shot MultiBox Detector), Faster-RCNN (Faster-Region Convolutional Neural Network), and unimproved YOLOv5s on the constructed pavement crack image dataset. The results show that compared with other algorithms, when mAP@0.5 (mean Average Precision when IoU=0.5) values are improved by 10.6%, 9.2%, and 4.6%, respectively, and cracks are identified better than the other three models.

INDEX TERMS Complex environments, deep learning, YOLOv5s, pavement crack detection.

I. INTRODUCTION

As the first part of the road structure to come into contact with external forces, the pavement is not only subjected to the loading pressure of vehicles but also affected by factors such as temperature changes, road corrosion, and human damage [1], [2]. Damage to road structures usually manifests itself first in the pavement, so the quality of the pavement structure directly affects the entire life cycle of a road project.

Pavement inspection is an integral part of the quality control of highway engineering construction, which is of great significance to the progress and cost control of highway engineering. When cracks appear in the pavement, if they are not dealt with in time, they will destroy the base layer, further expand the cracks, and even create breaks and potholes. This not only poses a threat to vehicle movement and causes traffic accidents but also increases the difficulty and cost of traffic regulation and transportation services [3]. Timely detection and repair of cracks prevent cracks from expanding and prolonging the service life of roads [4], [5]. Therefore, it is essential to study pavement crack detection technology.

This paper presents an enhanced YOLOv5s algorithm that significantly improves the accuracy and robustness of crack detection through three key strategies. First, the integration of the CBAM module strengthens the feature extraction capability for crack regions. Second, the adoption of the BiFPN structure effectively mitigates the issues of missed detections caused by variations in illumination and scale, while also enhancing the recognition accuracy of fine cracks. Finally, by incorporating a P6 layer and introducing a CA module in the detection head, the algorithm's multi-scale crack detection capability is further optimized, ensuring that the model maintains high precision and stability even in complex environments.

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II. RELATED WORKS

This section first outlines the traditional crack detection methods. Subsequently, a deep learning-based crack detection method is presented.

A. TRADITIONAL METHODS

Pavement cracks are a common problem in road maintenance, increasing the workload of pavement repair and maintenance. Accurately detecting these cracks plays a crucial role in suppressing their harm [6]. The current methods for detecting pavement cracks are mainly manual detection [7] and image processing techniques [8]. The traditional manual inspection needs to be managed by sealing and controlling the inspection roadway, considering the safety of the inspectors. However, this approach is inefficient, has a long cycle time, and is costly [9]. In addition, determining the type of crack is susceptible to subjective factors, resulting in poor accuracy. Therefore, improving the efficiency and accuracy of pavement crack detection in complex environments is crucial to preventing damage to road structures and traffic accidents. Pavement crack detection based on image processing has become a mainstream approach [10], [11], [12], [13], [14], and these algorithms can be divided into two categories: methods based on traditional image processing [15] and methods based on deep learning [16].

Traditional image processing methods have the advantages of low cost and high efficiency, and their accuracy gradually improves with the development of technology. Therefore, many researchers are trying to use computer vision to detect cracks in pavements [17], [18], [19], [20], [21]. Traditional road image processing techniques mostly rely on colour [22], texture [23] and geometric features [24] to segment out pavement defects and then classify and match them with classifiers to ultimately achieve the detection of pavement cracks. However, due to the complexity of the pavement environment and the influence of uneven illumination, traditional image processing methods are very sensitive to image noise and damage and need to extract artificially designed features [25], cannot meet the requirements of model generalization ability and robustness in practical applications, and cannot accurately identify the crack type.

B. DEEP LEARNING METHODS

Compared with traditional image processing methods, deep learning methods can learn the feature information of cracks more comprehensively and accurately and are also less susceptible to the influence of the external environment, leading to a decrease in recognition accuracy. Therefore, image processing technology based on deep learning theory has the characteristics of higher accuracy and faster detection speed in detecting pavement cracks, among other things. Zhan et al. [26] proposed using CNN for crack detection. Research shows that CNN in pavement crack detection is better than traditional image processing techniques, but there is leakage detection in complex environments. Girshick et al. [27] first proposed the RCNN algorithm, significantly improving the average accuracy. Subsequently, Girshick et al. [28]. proposed the Fast-RCNN algorithm, which added a pooling layer to the RCNN algorithm and added the images directly to the network for training. This avoided computational redundancy due to too many feature extraction frames, but the model's ability to generalize and robustness are not high. Maeda et al. [29] proposed the application of SSD network in road crack detection; SSD network belongs to the lightweight network, has fast detection speed, and requires a small dataset, but in the case of uneven illumination, lane line interference, etc, the SSD network exists in the case of multi-detection and misdetection. Xia et al. [30] proposed a pavement distress detection method based on deep convolutional neural networks to classify pavement distresses. Still, the situation is idealized and does not apply to complex road conditions. Nie et al. [31] proposed a convolutional neural network method for pavement crack detection using the Yolov3 algorithm to identify pavement cracks, which achieved good recognition accuracy in different types of pavement crack detection. Still, there is a leakage phenomenon for some fine cracks in multi-targets. Ma et al. [32] combined YOLOv3 with a PCGAN adversarial network and a median flow algorithm, which accelerated the detection speed while improving the accuracy to a certain extent. Still, the algorithm was ineffective in detecting tortoise cracks. Park et al. [33] used the YOLOv4 algorithm to identify potholes in road surfaces and achieved good results but did not study the identification of road cracks. Ren et al. [34] proposed YOLOv5 based on improving the generalized feature pyramid, but it is ineffective in detecting cracks in small targets. Ahmed [35] detected pavement defects using the YOLOv5 model. However, at mAP@0.5, only 58.9% could be achieved.

The application of deep learning-based target detection technology in pavement crack detection is very promising. However, there are still some difficulties, mainly: 1) the dataset's insufficient pavement crack types and the model's weak generalization ability [36]. 2) The road environment is susceptible to the influence of passing vehicles, roadside green facilities, lane lines, and shadows, resulting in uneven light intensity and creating difficulties for the target detection model. 3) Some pavement cracks are small and large in number, easily missed, and wrongly detected, reducing the detection accuracy of pavement cracks. To solve these problems, this paper proposes an improved algorithm based on YOLOv5s. Firstly, CBAM is introduced after the C3 module of the backbone network to enhance the model's attention to the crack region. Secondly, the BiFPN network structure is used in Neck to improve the multiscale feature fusion, reduce the detection leakage due to illumination and scale factors, and enhance the recognition of fine cracks. Finally, the P6 detection head is added to the head, and the CA module is introduced to improve the multi-scale crack detection capability. This makes the model in this paper highly accurate and robust in complex environments.

In this paper, the YOLOv5s model is improved to improve the detection accuracy and the detection capability for multi-target tasks in complex scenarios. Section I of this paper describes the importance of pavement crack detection techniques in existing engineering fields. Section II of this paper introduces the traditional and deep learning based pavement crack detection techniques are presented. Section III of this paper introduces the original network model of YOLOv5s. Section IV will detail the optimization made in the backbone network as well as Neck network and Head part of this paper. Section V presents the analysis of experimental results, convergence analysis, classification accuracy assessment, ablation experiments, and horizontal comparison experiments in complex scenarios for the three improvement points proposed in Section IV. Section VI summarizes our work and looks ahead.

III. YOLOv5s NETWORK MODEL

The YOLOv5 series contains four network models: YOLOv5s, YOLOv5l, YOLOv5m, and YOLOv5x. YOLOv5s is the most miniature network model among them, so the detection speed of YOLOv5s is the fastest. The YOLOv5S algorithm, due to its great flexibility, high accuracy, and speed, has been used in the field of defect detection widely and is a leader in the application of lightweight convolutional neural networks [37]. Therefore, we applied it to pavement defect detection. The structure of the YOLOv5s network is shown in Figure1.

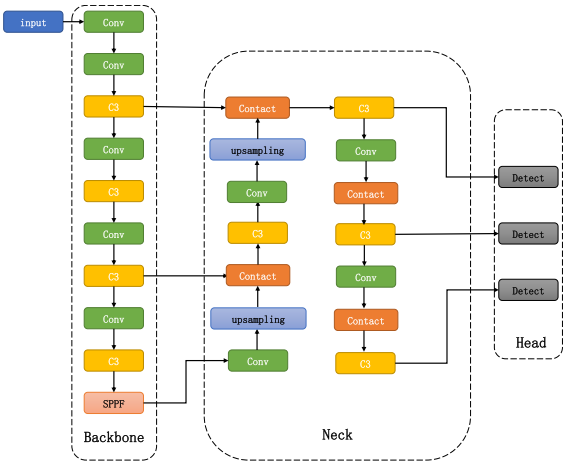


FIGURE 1. YOLOv5s network model.

YOLOv5s is composed of the Backbone, Neck network, and Head layer. When YOLOv5s handles the target detection task, the image is first input to the backbone network for feature extraction. The features extracted from the backbone network are inputted to the neck network for processing, and finally, the target category is output by the three detections of the head layer. Before the image is input to the backbone network, the Focus module slices the image as shown in Figure 2, which splits the high-resolution image into multiple low-resolution images, samples the images at intervals, and splices the sampling results in the channel dimension, which can expand the original number of channels by four times, which helps to improve the network speed. The backbone network uses CSPNet (Cross Stage Partial Network) with residual network structure [38], a lightweight network that can reduce the computation of the model by gradient changes while ensuring accuracy and reducing memory consumption.

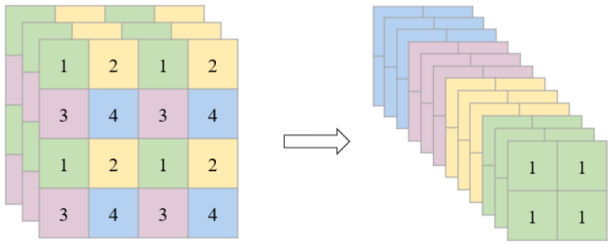


FIGURE 2. Slicing operations.

As the feature extraction goes deeper and deeper, some local information about the image will disappear. Using the neck network to fuse the feature maps of different network levels, we can obtain richer feature information for the image and input these processed features into the head layer to better classify and localize.

IV. IMPROVED NETWORK MODEL

Figure3 shows the improved model. CBAM is introduced after the C3 module to centralize the practical information of the complex scene and enhance the model's attention to the crack region. PANet in YOLOv5s is replaced with BiFPN to improve multi-scale feature fusion and thus reduce detection leakage due to illumination and scale factors. Finally, a P6 detection head for denser cracks was added to the Head, and a coordinate-aware CA module was introduced to improve multi-scale crack detection.

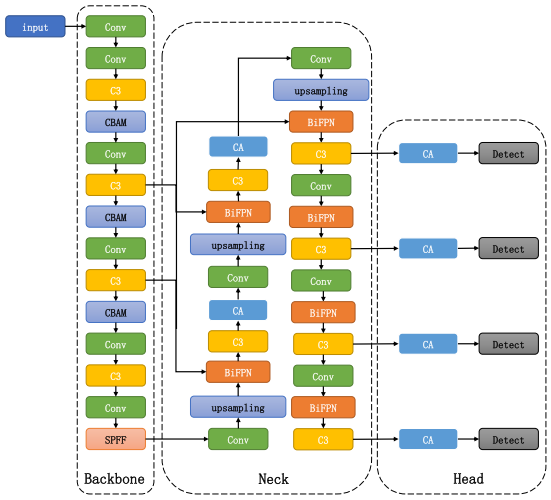


FIGURE 3. Algorithm framework of improved YOLOv5s.

A. BACKBONE NETWORK IMPROVEMENT

The CSPNet backbone of YOLOv5s is used for image feature extraction. After our extensive experiments, we found that

the link is not ideal for crack feature extraction. Then, after repeated upsampling and downsampling in the subsequent processing, there is information loss, and the fine cracks cannot be identified. The reasons for this are: the target of fine cracks is tiny, the features are not prominent enough, and they are easily disturbed by noise and background. For this reason, this paper introduces the CBAM [39] module in the backbone network of YOLOv5s to focus on the practical information in complex scenes and enhance the model's attention to the crack region.

As shown in Figure 4, CBAM contains a CAM (Channel Attention Module) and SAM (Spatial Attention Module), both of which focus on both spatial and channel dimensional features and utilize the spatial attention mechanism to help clarify the location of the tiny target, which uses both the Global Average Pooling and Global Maximum Pooling, the combination of these two forms of pooling can effectively prevent the loss of information, the use of channel attention mechanism can be obtained from the importance of each channel. For this thesis, the dataset has large-small cracks, dense arrangement, more noise, and other multiple interference situations; adding CBAM can make the network better by making the network focus on the location of the cracks, selectively ignoring some irrelevant information, focusing on the features that are useful to the target task, improving the detection accuracy, and enhancing the performance of the network.

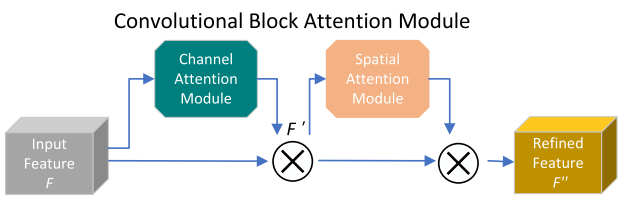


FIGURE 4. CBAM module.

The overall attention process can be summarized as follows:

$$F' = M_c(F) \otimes F \tag{1}$$

$$F'' = M_s(F') \otimes F' \tag{2}$$

where, F is the input feature map, F' is the output feature map of the CAM, F'' is the final refined output, $\otimes$ denotes element-wise multiplication.

Figure 5 shows the CAM module, which reduces the input feature map of $H \times W \times C$ to $1 \times 1 \times C$ by parallel global maximum pooling layer and global average pooling layer, and then the number of channels are compressed to $1/R$ (Reduction) times of the original one by the two-layer shared neural network module, and then restored to the original number of channels, and two activation results are obtained by the ReLU (Rectified Linear Unit) activation function. These two output results are summed element by element, and the sigmoid activation function is utilized to obtain the weight matrix of the input feature layer. Finally, it is multiplied with the input feature map to return to the size of $H \times W \times C$, and the input features required by the Spatial attention module are generated. The calculation formula is:

$$M_c(F) = \sigma\left(MLP\left(AvgPool\left(F\right)\right) + MLP(MaxPool\left(F\right))\right) \tag{3}$$

where, σ is the sigmoid activation function.

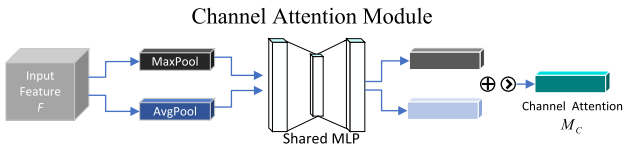


FIGURE 5. CAM structure diagram.

SAM takes the output feature map F' of the CAM as input and obtains two $H' \times W' \times 1$ feature maps by maximum pooling and average pooling, as shown in Figure 6, and then these two feature maps are channel spliced and transformed into 1-channel weight maps after 7×7 convolution operation, and generates the spatially compressed weight matrix M_s by sigmoid. Finally, M_s and F' of the module are multiplied to get the final generated features $H' \times W' \times 1$. The computational formula is:

$$M_S(F) = \sigma\left(f^{7\times7}\left([AvgPool\left(F\right); MaxPool\left(F\right)]\right)\right) \tag{4}$$

where, $f^{7\times7}$ is a 7×7 convolutional layer.

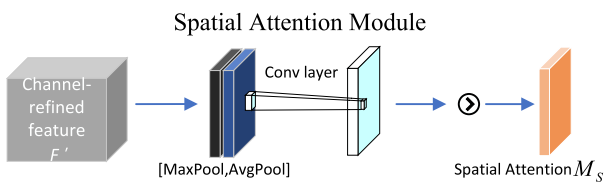


FIGURE 6. SAM structure diagram.

B. NECK NETWORK IMPROVEMENT

The Neck network of YOLOv5s uses the structure of PANet [40], as shown in Figure 7(a), which can effectively fuse the information for the pavement cracks; after our extensive experiments, we found that the original model has problems such as failing to identify the cracks. The reason is that the dataset of this paper has more types of cracks and an uneven distribution of cracks. For this reason, the original PANet network is improved to a BiFPN [41] network structure to enhance the feature extraction capability of the network model and eliminate the unilateral input nodes that contribute less to feature fusion. As shown in Figure 7(b), the BiFPN network enhances feature fusion at a higher level in the processing paths by processing each bi-directional path (top-down and bottom-up) as a feature network layer and repeating this process multiple times in the same layer. To further improve the detection efficiency of the deep network learning model, BiFPN uses separable

convolutional fusion features and adds bulk normalization and activation after each convolution. This design approach enhances the information transfer between feature maps at different network levels. It effectively improves the problem of loss of image feature information due to feature map resolution changes.

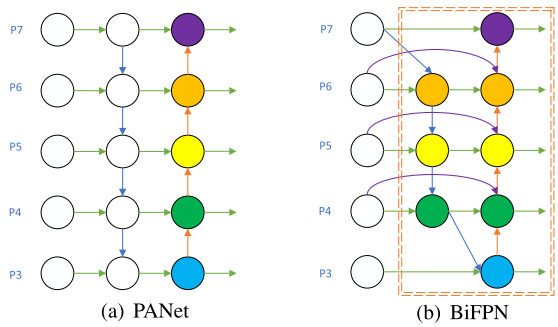


FIGURE 7. Network architecture diagram of PANet and BiFPN.

C. DETECTION HEAD PART IMPROVEMENT

The main body of the head part of YOLOv5s is 3 Detect detectors. The previously extracted features are utilized for prediction since cracks belong to small target detection. For some crack images taken by UAVs, as shown in the dataset of this paper, the small targets become even more challenging to recognize, so this paper improves the 3 detectors to 4 detectors and adds CA module [42] before each detector, aiming to improve the recognition rate for multi-target small target detection through the fusion of multiple detectors and CA module attention mechanism to improve the recognition rate of multi-target small target detection.

CA considers both inter-channel relationship and position information, as shown in Figure 8; it not only captures cross-channel information but also contains direction-aware and position-sensitive information; it can selectively ignore some irrelevant information, focus on the features that are useful for the target task, and make full use of the captured position information to locate the region of interest precisely. For a pre-trained model, especially for intensive crack target detection like this paper, the CA module will improve the accuracy of crack recognition. Increasing the detection head, on the one hand, increases the network's depth so that the network can better learn the multilevel feature information of the target and enhances the network's ability to detect multiscale targets in complex environments. On the other hand, it can pass more shallow features into deep features so that the network acquires more small target information and improves the model's ability to detect fine cracks.

In terms of capturing the spatial structure, given input X, we encode each channel using two pooling kernels $H \times 1$ and $W \times 1$ along the horizontal and vertical coordinates, respectively, and perform feature aggregation along the two directions, returning a pair of directionally-aware attentional features Z^h and Z^w. To achieve the purpose

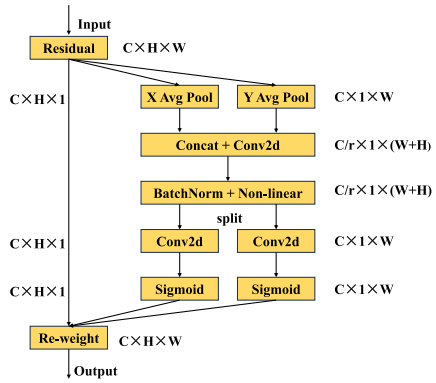


FIGURE 8. CA module.

of capturing remote dependencies along one spatial direction while retaining precise location information along another spatial direction. It helps the network to locate the target to be detected more accurately. The calculation formula is:

$$z_c^h(h) = \frac{1}{W} \sum_{0 \leqslant i \leqslant W} x_c(h, i) \tag{5}$$

$$z_c^w(w) = \frac{1}{H} \sum_{0 \leqslant i \leqslant H} x_c(j, w) \tag{6}$$

where, z_c^h and z_c^w represent the output feature maps of the cth channel with height h and width w, $(i, j)$are the pixels of the feature maps.

The relationship between each channel is efficiently processed based on fully utilizing the crack region extracted in the previous step. Attention generation is a spliced stack of perceptual feature maps in both directions, which are feature transformed by the shared 1×1 convolutional transform function F_1, BN, and nonlinear activation to obtain the feature mapping f as shown in equation (7):

$$f = \delta\left(F_1\left(\left[Z^h, Z^w\right]\right)\right) \tag{7}$$

where, $\left[Z^h, Z^w\right]$ denotes the spatial dimension splicing performed, δ is a nonlinear activation function.

Then f is decomposed in the spatial dimension into two separate tensors, $f^w \in R^{\frac{c}{r} \times W}$ and $f^h \in R^{\frac{c}{r} \times H}$ [43], with r denoting the downsampling ratio, which controls the size of the module, and then the feature maps f^h and f^w are transformed to the same channel as the input features using two (1×1) convolutional transform functions f^h and f^w. Then, the Sigmoid activation function is used to constrain the output values between (0, 1) to get two attention weights, g^h, and g^w, in the height and width directions, and the output of the CA module is shown in equation (10):

$$g^h = \sigma\left(F_h\left(f^h\right)\right) \tag{8}$$

$$g^w = \sigma\left(F_w\left(f^w\right)\right) \tag{9}$$

$$y_c(i, j) = x_c(i, j) * g_c^h(i) * g_c^w(j) \tag{10}$$

where, σ is the Sigmoid activation function.

V. PAVEMENT CRACK DETECTION EXPERIMENTS

To validate the effectiveness of the improved algorithm proposed in this paper for pavement crack detection, we conducted independent comparison experiments with three algorithms, SSD, YOLOv5s, and Faster-RCNN, on the dataset constructed in this paper, and further compare and evaluate the performance of the model proposed in this paper for pavement defect detection applications.

A. DETAILS

The dataset of this paper is obtained from the publicly available dataset [44] with multiple pavement cracks. The crack images contain various types of cracks, including Alligator crack, Longitudinal crack, Oblique crack, Transverse crack, Oblique crack, Pothole, and Repair, which belong to the common types of pavement cracks. To make full use of the dataset and improve the model's generalization ability, the image is reduced to a size of 640×640 so that the YOLO model can perform the best training performance. The dataset of 3152 images is allocated according to 9:1, 2837 as the training set, and 315 as a validation set [45].

The experimental environment and hyperparameter settings used in this experiment are shown in Table 1.

TABLE 1. Experimental environment and hyperparameter settings.

Name	Information	Parameter	Value
System	Windows10	Batch-size	16
GPU	NVIDIA RTX 3090	Epochs	500
Pytorch	2.0.1	Initial learning rate	0.01
CUDA	11.8	Momentum	0.937
CUDNN	8.1.0	weight_decay	0.0005
Python	3.10	Img_size	640×640

To verify the accuracy of the model proposed in this thesis, the metrics used in this paper are P (Precision), R (Recall), and mAP. The IoU selected for the model is 0.5; the confidence threshold for generating the prediction frame is 0.5; and the nms_iou used for non-great inhibition is 0.3. The calculation formula is:

$$P = \frac{TP}{TP + FP} \tag{11}$$

$$R = \frac{TP}{TP + FN} \tag{12}$$

$$AP = \frac{1}{11} \sum_{\gamma \epsilon \{0,0.1,0.2...1\}} \rho \quad interp\,(r) \tag{13}$$

$$mAP = \frac{1}{Q_R} \sum_{q \epsilon Q_R}^{q} AP\,(q) \tag{14}$$

where, TP is the number of crack images predicted correctly by positive samples; FP denotes the number of crack images mispredicted by negative samples; FN denotes the number of crack images mispredicted by positive samples; and AP denotes the average accuracy rate. Since there are many types of cracks, the recognition accuracy cannot be measured by referring only to the AP rate. However, the accuracy of all types of cracks can be achieved by obtaining the mAP rate, which is the average of the AP corresponding to all crack categories. The PR curve is plotted with R as the horizontal coordinate and P as the vertical coordinate. The larger the area under the model curve, the better the model's performance is proven.

B. EXPERIMENTAL RESULTS

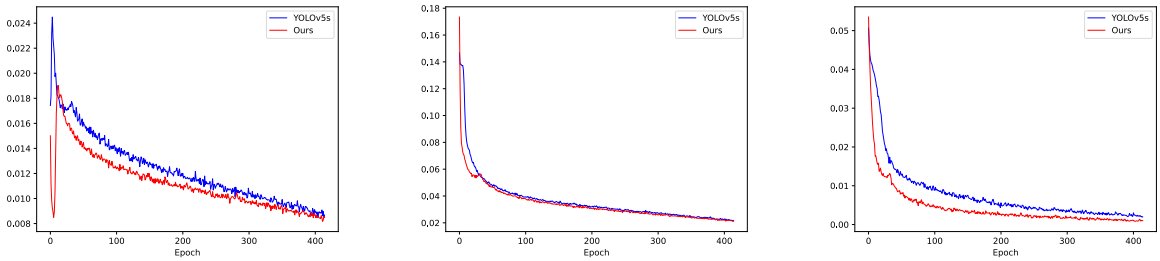
In this paper, the original YOLOv5s model and the improved model were trained on public datasets, and the mAP rates and loss function plots for recognizing different types of cracks were obtained. The results show that the improved algorithm is more effective in classifying and recognizing various types of cracks. Compared with the original algorithm, the improved model shows the feasibility of training with large fluctuations in loss values at the beginning of training. As training proceeds, the target confidence loss (train_obj_loss) gradually decreases and converges more quickly, indicating that the algorithm has higher accuracy in distinguishing different crack categories. The decrease in the box regression loss (train_box_loss) indicates that the model can accurately predict the location of the target cracks. The comparison of the category classification loss (train_cls_loss) shows that the model in this paper has a significant decreasing trend and faster convergence, which indicates that the improved model can recognize the target in the image more effectively. The gradual convergence of the loss values signifies the robustness of the model and its effectiveness in training on a given dataset.

However, in this experiment, our improved model based on YOLOv5s employs an early stopping strategy: when the model performance is no longer improving on the validation set during training, the early stopping strategy terminates the training early, which saves training time and computational resources, improves the model generalization ability, and prevents the model from overfitting, and the comparison of the training loss functions is shown in Figure 9.

We reproduce the improved algorithm and the original algorithm on the test dataset to evaluate the performance of the model and obtain the PR curve, as shown in Figure 10. The more significant the area enclosed by the PR curve and the coordinates, the higher the AP value, and the better the performance of the model. From Figure 10, it can be seen that the performance of the improved model in this paper is significantly better than the original model performance.

1) COMPARISON OF FOUR MODEL INDICATORS

Table 2 shows that this paper's algorithm is not the highest in P and R, but compared to other algorithms mAP@0.5 values are improved by 10.6%, 9.2%, 4.6% respectively, the precision is improved by 31.1% and 7.8% compared to Faster-RCNN, YOLOv5s respectively, and it is reduced by 1.4% compared to SSD model. The FPS data shows that the improved model meets the real-time requirements, and the



(a) train_obj_loss (b) train_box_loss (c) train_cls_loss

FIGURE 9. Training loss function graphs.

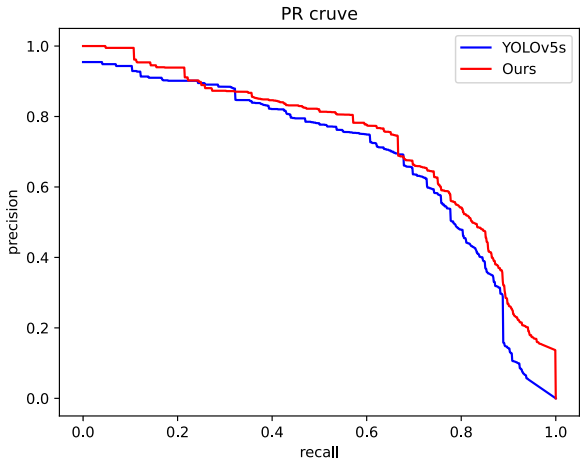


FIGURE 10. Comparison of PR cruve.

TABLE 2. Test results of different detection models (Accuracy, recall for each category is provided and averaged through independent test set testing).

Model	P	R	mAP@0.5	FPS
SSD	75.4%	49.9%	62.4%	57
Faster-RCNN	42.9%	75.3%	63.8%	5
YOLOv5s	66.2%	67.0%	68.4%	156
Ours	74.0%	71.0%	73.0%	123

real-time processing capability is slightly reduced compared with the original model due to the increased modules of the improved model, and the performance of this paper's model is significantly better than that of the three models mentioned above with the loss of a smaller frame rate [46].

2) ABLATION EXPERIMENTS

Through Table 3, the effect of different improved structures on the pavement crack image detection results, we found that the mAP@0.5 value by fusing CBAM, CA, BiFPN, and P6 detection head is the highest, which proves that our work is practical. All four improvement strategies can increase the model's performance, which can be used as a reference for other work in the field of defect detection. In conclusion, our proposed improved YOLOv5s deep network learning model

TABLE 3. Effect of different improved structures on the image detection results of pavement cracks.

BiFPN	CBAM	CA	P6	P	R	mAP@0.5
✓				76.3%	60.0%	69.1%
	✓			73.8%	63.1%	69.3%
		✓		65.6%	73.3%	69.7%
			✓	76.2%	67.7%	71.5%
✓			✓	73.8%	63.2%	71.7%
✓		✓	✓	70.1%	68.5%	71.5%
✓	✓	✓	✓	74.0%	71.0%	73.0%

far exceeds the remaining three current network models regarding mAP@0.5, P, and R metrics, and it represents a lightweight target detection model.

3) TEST RESULTS OF DIFFERENT NETWORK MODELS IN COMPLEX ENVIRONMENTS

In this experiment, different network models are selected for experimental testing in complex environments (weather, light shadows, lane lines, green belts, and other conditions) [47].

As shown in Figure 11(a), we tested four models in the small cracks and obstacles interference situation of pavement detection and compared them with the ground truth; from the figure, it can be seen that the SSD detection effect of the worst appeared in the case of missed detection, Faster-RCNN and YOLOv5s appeared in the effect of repeated detection, this paper's algorithm is the closest to the ground truth. As shown in Figure 11(b), under the shadow interference, SSD has missed detection, Faster-RCNN has detection error, YOLOv5s and this paper's algorithm are better, as shown in Figure 11(c), under the lane line interference and multi-target detection, Faster-RCNN has the duplicate missed detection as SSD, and YOLOv5s has the effect of repeated detection, this paper's algorithm is almost the same as the ground truth. The algorithm is almost consistent with the ground truth. As shown in Figure 11(d) under the task of large green belt interference as well as small target detection, YOLOv5s has missed detection, and Faster-RCNN and SSD have different degrees of duplicate detection, while the algorithm in this paper is consistent with the ground truth.

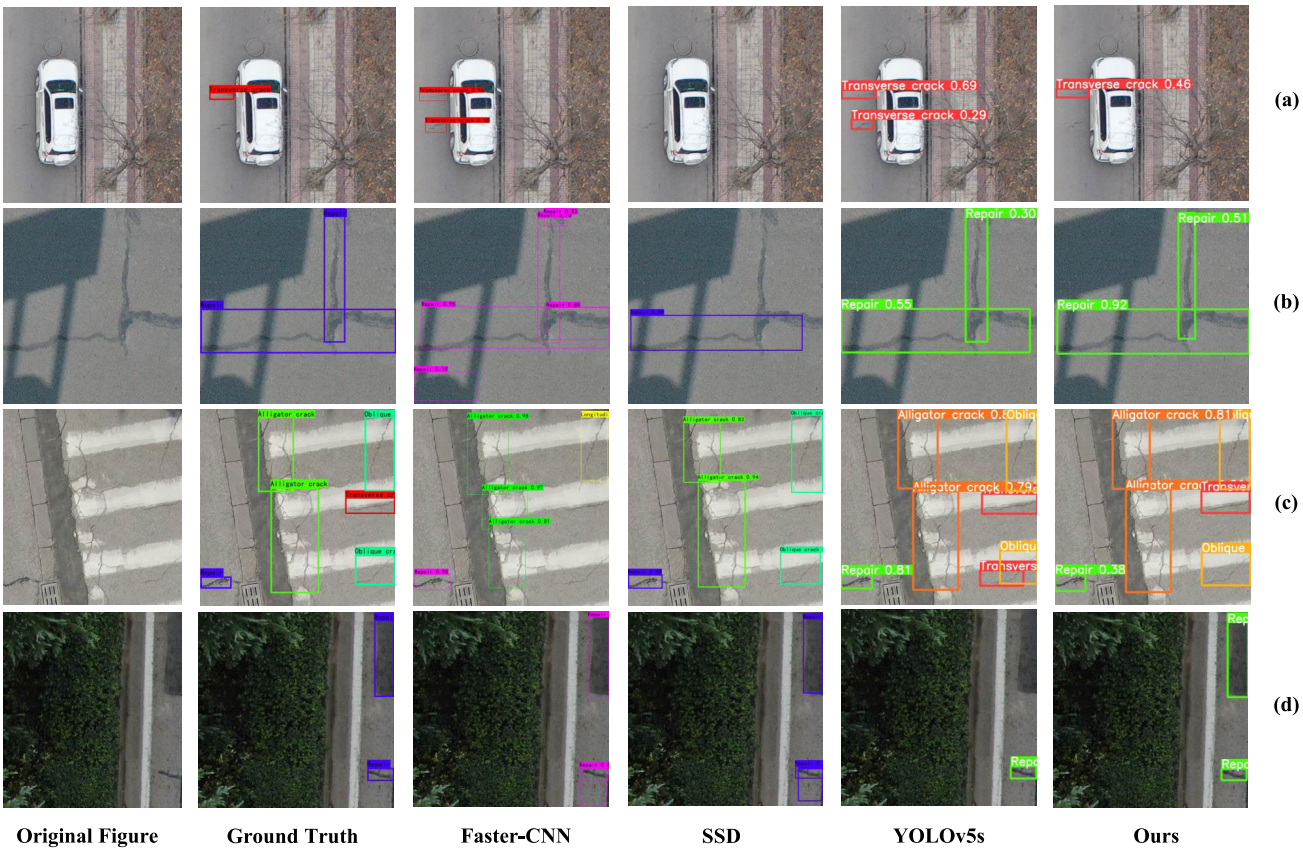


FIGURE 11. Test results of different network models in complex environments. (a) Model detection results under small cracks with obstacle interference. (b) Model detection results under shadow interference. (c) Model detection results under lane interference and multi-target detection tasks. (d) Model detection results under a large green belt and small target detection environment.

VI. CONCLUSION

The detection of pavement cracks is an area of significant research interest. This paper proposes an enhanced YOLOv5s model to improve accuracy and mitigate the effects of external factors. The integration of CA and CBAM modules allows the model to utilize comprehensive scene information from the dataset, considering both inter-channel relationships and location details. This enables more precise identification of target areas, enhancing performance in detecting pavement cracks. By replacing the BiFPN network, the model fuses more location and detail information, enhancing its focus on small target areas and improving multi-target detection through multiple detection heads. Compared to three other detection algorithms, the improved YOLOv5s show superior anti-interference capability, higher sensitivity to small targets, lower miss detection rates, and better generalization and robustness. The improved model and preprocessing operation in this paper can only extract the location and classification information of pavement cracks, without quantitatively analyzing the severity of the crack lesions. Future research will focus on assessing the severity of the cracks and using image segmentation methods for further information extraction. Given the YOLOv5s network model's excellent running speed, compact size, and ease of deployment, the optimized model can be further lightweight for better size and speed, making it suitable for embedded devices such as cars or drones to enhance the automatic detection of pavement defects.

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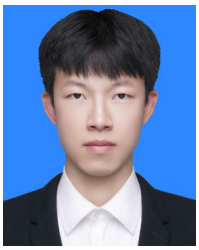

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